The Hilbert-Space Quadrature Conjecture: Time as the Wave-Function-Generated Quadrature of Circular Phase

Steven Michael Allen Owens *Independent Researcher* (Dated: October 17, 2026)

Abstract

We formulate and prove the conditional consistency of the *Hilbert-Space Quadrature Conjecture*, which proposes that physical time is not a primitive external background parameter, but an emergent, state-generated observable arising from the quadratic mapping ("squaring") of the intrinsic circular U(1) phase fiber of Hilbert space. Let \mathcal{H}_U = \mathcal{H}_C \otimes \mathcal{H}_S be the bipartite Hilbert space of a closed, globally stationary quantum universe satisfying a Wheeler–DeWitt-type constraint \widehat{C}\vert{}\Psi\rangle = 0. To respect Pauli's theorem for lower-bounded clock Hamiltonians, primitive clock states live on the circle S^1 and resolve the identity via a covariant Holevo phase positive-operator-valued measure (POVM) E(d\theta). We prove three main results: (1) the apparent time-evolving system wave function \vert{}\psi(\theta)\rangle = \langle\theta\vert{}\Psi\rangle is the conditional slice of the universal state, and observable temporal order is generated by the state-dependent joint quadrature map Q_\Psi(\theta, a) = \vert{}\langle\theta, a\vert{}\Psi\rangle\vert{}^2; (2) in the sharp-phase limit, conditional quadrature evolution reproduces the standard unitary Schrödinger equation; and (3) for physical clocks with a semi-bounded spectrum and finite energy spread \Delta E_C, the quadrature map induces an exact Lindblad double-commutator master equation governing intrinsic temporal dephasing, while connecting directly to the Connes–Rovelli thermal time modular flow \Delta_\Psi^{is}.

I. Introduction: The Problem of External Time and the Phase Fiber

In standard non-relativistic quantum mechanics, time t \in \mathbb{R} enters the Schrödinger equation as a classical, external parameter rather than a quantum observable on the Hilbert space \mathcal{H}. As demonstrated by Pauli [1], if the system Hamiltonian \widehat{H} is semi-bounded below (\operatorname{Spec}(\widehat{H}) \subset [E_0, \infty)), there can exist no self-adjoint time operator \widehat{T} satisfying the canonical Weyl commutation relation [\widehat{T}, \widehat{H}] = i\hbar I. Conversely, in canonical quantum gravity and reparameterization-invariant cosmology, the Wheeler–DeWitt equation \widehat{C}\vert{}\Psi\rangle = 0 eliminates external time altogether, leaving a "frozen" universal state \vert{}\Psi\rangle [2].

Two major programs have sought to recover temporal dynamics from timeless quantum states: the **Page–Wootters conditional probability mechanism** [3, 4], which extracts relational

evolution from entanglement between a quantum clock $\mathcal{H}_C$ and a system $\mathcal{H}_S$, and the **Connes–Rovelli thermal time hypothesis** [5], which identifies physical time flow with the state-dependent Tomita–Takesaki modular automorphism group σ_s^Ψ of a faithful density state.

In this paper, we unify and extend these frameworks through a geometric and operational principle we term the **Hilbert-Space Quadrature Conjecture**. Our starting observation is geometric: complex Hilbert space $\mathcal{H}$ possesses an intrinsic circular phase structure given by the principal U(1) bundle over projective ray space,

$$U(1) \hookrightarrow S(\mathcal{H}) \xrightarrow{\pi} \mathbb{P}(\mathcal{H}), \qquad S(\mathcal{H}) = \{\psi \in \mathcal{H} : \Vert{}\psi\Vert{} = 1\}.$$

Raw phase angles $\theta \in S^1 \cong U(1)$ along a single fiber are gauge-redundant and unobservable. However, when a universal wave function $\vert{}\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ entangles a semi-bounded clock sector with a dynamical system, the Born-rule quadratic map—and its phase-space generalization, the Wigner–Weyl distribution [6]—"squares" the circular S^1 phase amplitudes of $\vert{}\Psi\rangle$ into a real, positive probability measure over clock readings. In this formulation, *time is not an arena in which the wave function evolves; rather, the wave function of the clock sector is the physical generator that squares the circular phase of Hilbert space into observable temporal order.*

II. Mathematical Setting and the Covariant Clock POVM on S^1

Let the universal Hilbert space be the tensor product $\mathcal{H}_U = \mathcal{H}_C \otimes \mathcal{H}_S$, where $\mathcal{H}_C$ is the clock Hilbert space and $\mathcal{H}_S$ is the system Hilbert space.

To ensure the clock is physically realizable with a ground state, we require its number (or energy-ladder) generator $\widehat{N}$ to be lower-bounded with spectrum $\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0 = \{0, 1, 2, \dots\}$ and orthonormal eigenstates $\{\vert{}n\rangle_{C}\}_{n=0}^\infty$. Thus, $\mathcal{H}_C$ is isomorphic to the Hardy space $H^2(S^1) \subset L^2(S^1, d\theta/2\pi)$. The clock Hamiltonian is given by $\widehat{H}_C = \hbar \omega \widehat{N}$, where $\omega > 0$ is the fundamental clock frequency scale. Because $\operatorname{Spec}(\widehat{N})$ is bounded below by 0, Pauli's theorem [1] forbids a self-adjoint phase/time operator on $\mathcal{H}_C$. Following Holevo [7], we instead construct the family of generalized circular phase states $\vert{}\theta\rangle \in \Phi_C^\times$ (where $\Phi_C \subset \mathcal{H}_C \subset \Phi_C^\times$ is the Gel'fand triple of rapidly decaying Fourier coefficients) for $\theta \in [0, 2\pi) \cong S^1$:

$$\vert{}\theta\rangle = \sum_{n=0}^{\infty} e^{in\theta} \vert{}n\rangle.$$

Note that because the sum begins at n=0 rather than -\infty, the inner product between distinct phase states is non-orthogonal and given by the boundary Szegő kernel:

$$\langle \theta \vert{} \theta' \rangle = \sum_{n=0}^{\infty} e^{-in(\theta - \theta')} = \pi \delta_{2\pi}(\theta - \theta') + \frac{1}{1 - e^{i(\theta - \theta' + i0^+)}}.$$

Proposition 1 (Covariant Circular Clock POVM)

The operator-valued measure $E(d\theta)$ on the Borel \sigma-algebra \mathcal{B}(S^1) defined by

$$E(d\theta) = \vert{}\theta\rangle\langle\theta\vert{} \, \frac{d\theta}{2\pi}$$

forms a normalized, shift-covariant positive-operator-valued measure (POVM) on \mathcal{H}_C satisfying:

1. **Positivity:** \langle \phi \vert{} E(\Delta) \vert{} \phi \rangle = \int_\Delta \frac{d\theta}{2\pi} \vert{}\langle \theta \vert{} \phi \rangle\vert{}^2 \ge 0 *for all* \Delta \in \mathcal{B}(S^1) *and* \vert{}\phi\rangle \in \mathcal{H}_C.
2. **Resolution of the Identity:** \int_0^{2\pi} E(d\theta) = I_C.
3. **Phase Covariance:** e^{-i\phi \widehat{N}} E(\Delta) e^{i\phi \widehat{N}} = E(\Delta \oplus \phi) *for all* \phi \in S^1, *where* \oplus *denotes addition modulo* 2\pi.

Proof

Positivity follows immediately from the positive-semidefiniteness of the rank-one outer product \vert{}\theta\rangle\langle\theta\vert{}. To verify the resolution of the identity on the dense domain \Phi_C \subset \mathcal{H}_C, expand the matrix elements in the \{\vert{}n\rangle\} basis:
\int_0^{2\pi} \frac{d\theta}{2\pi} \vert{}\theta\rangle\langle\theta\vert{} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \left(\int_0^{2\pi} \frac{d\theta}{2\pi} e^{i(n-m)\theta} \right) \vert{}n\rangle\langle m\vert{} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \delta_{nm} \vert{}n\rangle\langle m\vert{} = \sum_{n=0}^{\infty} \vert{}n\rangle\langle n\vert{} = I_C.
For phase covariance, apply the unitary clock shift operator U(\phi) = e^{-i\phi \widehat{N}} to \vert{}\theta\rangle:
U(\phi)\vert{}\theta\rangle = \sum_{n=0}^{\infty} e^{-in\phi} e^{in\theta} \vert{}n\rangle = \sum_{n=0}^{\infty} e^{in(\theta - \phi)} \vert{}n\rangle = \vert{}\theta - \phi\rangle.
Integrating over a measurable subset \Delta \subset S^1 with the substitution \theta' = \theta - \phi \pmod{2\pi} yields U(\phi) E(\Delta) U^\dagger(\phi) = E(\Delta \oplus \phi). \blacksquare

III. The Quadrature Conjecture and Conditional Schrödinger Dynamics

We now state the formal core of the framework.

Definition 1 (The Quadrature Map)

Let \vert{}\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S be a physical state of the closed composite universe, and let \vert{}\theta\rangle\langle\theta\vert{} \frac{d\theta}{2\pi} be the covariant clock POVM of Proposition 1. The **unnormalized conditional system wave function** at clock phase \theta \in S^1 is the partial projection
\vert{}\Psi(\theta)\rangle := (\langle \theta \vert{}_C \otimes I_S)\vert{}\Psi\rangle \in \mathcal{H}_S.
The **Clock Quadrature Distribution** Q_\Psi : S^1 \to \mathbb{R}_{\ge 0} and the **Joint System-Clock Quadrature** Q_\Psi(\theta, a) for any system observable projector \Pi_a = \vert{}a\rangle\langle a\vert{}_S are defined as the quadratic mappings:
Q_\Psi(\theta) := \Vert{}\Psi(\theta)\Vert{}_{\mathcal{H}_S}^2 = \langle \Psi \vert{} \left(\vert{}\theta\rangle\langle\theta\vert{}_C \otimes I_S\right) \vert{} \Psi \rangle,
Q_\Psi(\theta, a) := \vert{}\langle \theta, a \vert{} \Psi \rangle\vert{}^2 = \langle \Psi \vert{} \left(\vert{}\theta\rangle\langle\theta\vert{}_C \otimes \Pi_a\right) \vert{} \Psi \rangle.

Conjecture 1 (The Hilbert-Space Quadrature Conjecture)

In any closed quantum universe $\mathcal{H}_U = \mathcal{H}_C \otimes \mathcal{H}_S$ *governed by a timeless constraint* $\widehat{C}\vert{}\Psi\rangle = 0$*, there exists no external background time parameter. All observable temporal ordering and dynamical evolution are generated internally by the universal wave function* $\vert{}\Psi\rangle$ *through the quadratic map* $\Psi \mapsto Q_\Psi(\theta, a)$ *over the circular phase manifold* S^1*. The normalized conditional system state at clock reading* θ*,*

$$\rho_S(\theta) = \frac{\operatorname{Tr}_C\left[(\vert{}\theta\rangle\langle\theta\vert{}_C \otimes I_S)\vert{}\Psi\rangle\langle\Psi\vert{}\right]}{Q_\Psi(\theta)} = \frac{\vert{}\Psi(\theta)\rangle\langle\Psi(\theta)\vert{}}{\Vert{}\Psi(\theta)\Vert{}^2},$$

constitutes the complete physical state of the system at emergent time $t = \theta / \omega$*.*

Theorem 1 (Emergence of Conditional Schrödinger Evolution)

Let $\widehat{H}_S$ *be a self-adjoint system Hamiltonian on* $\mathcal{H}_S$ *with discrete eigenbasis* $\widehat{H}_S \vert{}E_k\rangle = E_k \vert{}E_k\rangle$*. Suppose the system spectrum is shifted and commensurate with the clock frequency scale* ω *such that* $E_k = \hbar \omega n_k$ *for* $n_k \in \mathbb{N}_0$*, and let* $\vert{}\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ *satisfy the global constraint*

$$\widehat{C}\vert{}\Psi\rangle = \left(\hbar \omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)\vert{}\Psi\rangle = 0.$$

Then, defining the emergent time parameter $t := \theta / \omega \in [0, 2\pi/\omega)$*, the conditional system state* $\vert{}\psi(t)\rangle := \vert{}\Psi(\omega t)\rangle / \sqrt{Q_\Psi(\omega t)}$ *satisfies the exact unitary Schrödinger equation:*

$$i\hbar \frac{\partial}{\partial t} \vert{}\psi(t)\rangle = \widehat{H}_S \vert{}\psi(t)\rangle,$$

and the marginal clock quadrature density $Q_\Psi(\theta)$ *is globally stationary across the circle* S^1*.*

Proof

Expand the general state $\vert{}\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ in the joint eigenbasis $\{\vert{}n\rangle_C \otimes \vert{}E_k\rangle_S\}$:

$$\vert{}\Psi\rangle = \sum_{n=0}^{\infty} \sum_{k} c_{n,k} \vert{}n\rangle_C \otimes \vert{}E_k\rangle_S.$$

Applying the constraint operator $\widehat{C}$ of Eq. (11) yields

$$\widehat{C}\vert{}\Psi\rangle = \sum_{n=0}^{\infty} \sum_{k} c_{n,k} (\hbar \omega n - E_k) \vert{}n\rangle_C \otimes \vert{}E_k\rangle_S = 0.$$

By linear independence of the basis vectors, $c_{n,k}$ vanishes unless $\hbar \omega n = E_k$, which enforces $n = n_k = E_k / (\hbar \omega) \in \mathbb{N}_0$. Writing $c_{n_k, k} = \alpha_k$ with normalization $\sum_k \vert{}\alpha_k\vert{}^2 = 1$, the stationary universal state takes the entangled form

$$\vert{}\Psi\rangle = \sum_{k} \alpha_k \vert{}n_k\rangle_C \otimes \vert{}E_k\rangle_S.$$

Projecting onto the covariant clock state $\langle \theta \vert{}_C = \sum_{m=0}^{\infty} e^{-im\theta} \langle m \vert{}_C$ via Definition 1 gives the conditional amplitude

$$\vert{}\Psi(\theta)\rangle = (\langle \theta \vert{}_C \otimes I_S)\vert{}\Psi\rangle = \sum_{k} \alpha_k e^{-i n_k \theta} \vert{}E_k\rangle_S = \sum_{k} \alpha_k e^{-i E_k \theta /$$

(\hbar\omega)} \vert{}E_k\rangle_S.
Differentiating Eq. (16) with respect to \theta and substituting t = \theta / \omega (\frac{\partial}{\partial \theta} = \frac{1}{\omega} \frac{\partial}{\partial t}) yields
i\hbar \frac{\partial}{\partial t} \vert{}\Psi(\omega t)\rangle = i\hbar \omega \frac{\partial}{\partial \theta} \sum_{k} \alpha_k e^{-i E_k \theta / (\hbar\omega)} \vert{}E_k\rangle_S = \sum_{k} \alpha_k E_k e^{-i E_k t / \hbar} \vert{}E_k\rangle_S = \widehat{H}_S \vert{}\Psi(\omega t)\rangle.
Finally, evaluating the clock quadrature Q_\Psi(\theta) via Eq. (8) gives
Q_\Psi(\theta) = \langle \Psi(\theta) \vert{} \Psi(\theta) \rangle = \sum_{k, j} \alpha_j^* \alpha_k e^{i(E_j - E_k)\theta/(\hbar\omega)} \langle E_j \vert{} E_k \rangle_S = \sum_{k} \vert{}\alpha_k\vert{}^2 = 1.
Because Q_\Psi(\theta) = 1 is independent of \theta, \vert{}\psi(t)\rangle = \vert{}\Psi(\omega t)\rangle is normalized for all t and satisfies Eq. (12). \blacksquare

IV. Finite-Clock Corrections and Intrinsic Lindblad Dephasing

In a realistic physical universe, either the clock has a finite energy spread \Delta E_C around a mean energy \bar{E}_C, or the system energies E_k are not exact integer multiples of \hbar\omega. In this regime, the clock state cannot be an infinitely sharp phase eigenstate; rather, conditioning on a physical clock reading \theta corresponds to projecting onto a smeared Gaussian clock wavepacket \vert{}\tau_\sigma(\theta)\rangle of phase variance \sigma_\theta^2 \sim (\hbar \omega / 2\Delta E_C)^2.

Theorem 2 (Finite-Clock Master Equation)

Suppose the clock readout POVM has finite temporal resolution characterized by a Gaussian phase uncertainty \sigma_t = \sigma_\theta / \omega = \sqrt{2\gamma_{\text{clock}}} \, \delta t over a coarse-grained time step \delta t, such that the effective conditional system density operator is given by the quadrature convolution
\bar{\rho}_S(t) = \int_{-\infty}^{\infty} \frac{ds}{\sqrt{2\pi \sigma_t^2}} \exp\left(-\frac{(s - t)^2}{2\sigma_t^2}\right) e^{-i\widehat{H}_S s/\hbar} \rho_S(0) e^{i\widehat{H}_S s/\hbar}.
Then \bar{\rho}_S(t) satisfies the exact Lindblad double-commutator master equation:
\frac{d\bar{\rho}_S(t)}{dt} = -\frac{i}{\hbar}\left[\widehat{H}_S, \bar{\rho}_S(t)\right] - \frac{\gamma_{\text{clock}}}{\hbar^2} \left[\widehat{H}_S, \left[\widehat{H}_S, \bar{\rho}_S(t)\right]\right],
where \gamma_{\text{clock}} = \frac{\hbar^2}{4 (\Delta E_C)^2 \tau_c} (with \tau_c the clock correlation scale) is the intrinsic clock-induced dephasing rate.

Proof

Write \rho_S(0) = \sum_{j,k} \rho_{jk}(0) \vert{}E_j\rangle\langle E_k\vert{} in the eigenbasis of \widehat{H}_S. Substituting into Eq. (19) and setting \omega_{jk} = (E_j - E_k)/\hbar, we evaluate the Gaussian Fourier integral:
\bar{\rho}_{jk}(t) = \rho_{jk}(0) \int_{-\infty}^{\infty} \frac{ds}{\sqrt{2\pi \sigma_t^2}} \exp\left(-\frac{(s - t)^2}{2\sigma_t^2} - i\omega_{jk} s\right) = \rho_{jk}(0) e^{-i\omega_{jk} t} \exp\left(-\frac{1}{2} \omega_{jk}^2 \sigma_t^2\right).

Substituting the diffusive clock variance scaling $\sigma_t^2(t) = 2 \gamma_{\text{clock}} t$ (which characterizes a quantum clock whose phase variance grows linearly over multiple ticks or possesses an intrinsic Markovian resolution floor γ_{clock} [8]) gives $\bar{\rho}_{jk}(t) = \rho_{jk}(0) \exp\left(-i\frac{E_j - E_k}{\hbar}t - \frac{(E_j - E_k)^2}{\hbar^2} \gamma_{\text{clock}} t\right)$.
Differentiating Eq. (22) with respect to t yields
$\frac{d\bar{\rho}_{jk}(t)}{dt} = \left[-\frac{i}{\hbar}(E_j - E_k) - \frac{\gamma_{\text{clock}}}{\hbar^2}(E_j - E_k)^2 \right] \bar{\rho}_{jk}(t)$.
Rewriting the eigenvalue factors $(E_j - E_k)$ and $(E_j - E_k)^2$ in operator form via the commutator $[\widehat{H}_S, \vert{}E_j\rangle\langle E_k\vert{}] = (E_j - E_k)\vert{}E_j\rangle\langle E_k\vert{}$ and double commutator $[\widehat{H}_S, [\widehat{H}_S, \vert{}E_j\rangle\langle E_k\vert{}]] = (E_j - E_k)^2 \vert{}E_j\rangle\langle E_k\vert{}$ immediately yields Eq. (20). $\blacksquare$

V. Equivalence to Modular Flow and Wigner Phase-Space Quadrature

Finally, we connect the Quadrature Conjecture to the Connes–Rovelli thermal time hypothesis [5] and Wigner–Weyl phase space [6].

Proposition 2 (Modular Time as Clock-Sector Quadrature)

Let $\vert{}\Psi\rangle = \sum_k \alpha_k \vert{}n_k\rangle_C \otimes \vert{}E_k\rangle_S$ be a globally stationary entangled state with non-vanishing Schmidt coefficients $\alpha_k > 0$. Then the reduced clock state $\rho_C = \operatorname{Tr}_S(\vert{}\Psi\rangle\langle\Psi\vert{})$ defines a Tomita–Takesaki modular operator $\Delta_\Psi = \rho_C \otimes \rho_S^{-1}$ whose modular flow $\sigma_s^\Psi(A) = \Delta_\Psi^{is} A \Delta_\Psi^{-is}$ coincides with the unitary Quadrature Time evolution at the inverse-temperature Boltzmann distribution $\vert{}\alpha_k\vert{}^2 = Z^{-1} e^{-\beta E_k}$, with $t = -\hbar \beta s$.

Proof

Tracing out $\mathcal{H}_S$ from Eq. (15) yields the diagonal reduced clock density matrix $\rho_C = \sum_k \vert{}\alpha_k\vert{}^2 \vert{}n_k\rangle\langle n_k\vert{}_C$, and tracing out $\mathcal{H}_C$ yields $\rho_S = \sum_k \vert{}\alpha_k\vert{}^2 \vert{}E_k\rangle\langle E_k\vert{}_S$. On the support subspace where $\alpha_k > 0$, ρ_S is faithful, and its modular Hamiltonian is $K_S = -\ln \rho_S = -\sum_k (\ln \vert{}\alpha_k\vert{}^2) \vert{}E_k\rangle\langle E_k\vert{}_S$. For a Gibbs distribution $\vert{}\alpha_k\vert{}^2 = Z^{-1} e^{-\beta E_k}$, we have $K_S = \beta \widehat{H}_S + (\ln Z) I_S$. Consequently, the modular automorphism group acting on any system observable $A_S \in \mathcal{B}(\mathcal{H}_S)$ is
$\sigma_s^\Psi(A_S) = \rho_S^{is} A_S \rho_S^{-is} = e^{-is K_S} A_S e^{is K_S} = e^{-is\beta \widehat{H}_S} A_S e^{is\beta \widehat{H}_S}$.
Comparing Eq. (24) with the Heisenberg evolution generated by Theorem 1, $A_S(t) = e^{i\widehat{H}_S t/\hbar} A_S e^{-i\widehat{H}_S t/\hbar}$, establishes exact equivalence under the identification $t = -\hbar \beta s$. $\blacksquare$

Connection to Wigner–Weyl Phase Space ("Squaring the Circle")

In the position representation of \mathcal{H}_S, the conditional wave function \psi(q, \theta) = (\langle \theta \vert{}_C \otimes \langle q \vert{}_S)\vert{}\Psi\rangle is mapped into the classical Cartesian phase space \mathbb{R}^{2n} via the quadratic Wigner transform [6]:
W_\Psi(q, p; \theta) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dy \, e^{-ipy/\hbar} \, \psi\left(q + \frac{y}{2}, \theta\right) \psi^*\left(q - \frac{y}{2}, \theta\right).
Applying Theorem 1 to Eq. (25) shows that the circular phase derivative \omega \frac{\partial}{\partial \theta} drives the Moyal bracket evolution \omega \frac{\partial W_\Psi}{\partial \theta} = \{\widehat{H}_S, W_\Psi\}_{\text{Moyal}}. Thus, the universal wave function \vert{}\Psi\rangle performs a dual quadratic operation: it maps the circular clock fiber S^1_C into the temporal measure Q_\Psi(\theta) while simultaneously mapping the system Hilbert space into the symplectic (q,p) square of classical mechanics.

VI. Conclusion and Empirical Outlook

The Hilbert-Space Quadrature Conjecture resolves the conceptual dichotomy between timeless universal wave functions and dynamical time evolution. By identifying primitive time with the circular phase fiber S^1 of a lower-bounded clock sector and observable time with the quadratic map Q_\Psi(\theta, a) = \vert{}\langle\theta, a\vert{}\Psi\rangle\vert{}^2, standard Schrödinger dynamics emerges conditionally (Theorem 1), while finite clock resources predict a testable Lindblad energy-dephasing signature \gamma_{\text{clock}} \propto (\Delta E_C)^{-2} (Theorem 2) accessible in modern trapped-ion, cavity-QED, and interferometric Page–Wootters platforms [9].

References

1. W. Pauli, *Die allgemeinen Prinzipien der Wellenmechanik*, in Handbuch der Physik, Vol. 24 (Springer, Berlin, 1933), p. 83.
2. B. S. DeWitt, *Quantum Theory of Gravity. I. The Canonical Theory*, Phys. Rev. **160**, 1113 (1967).
3. D. N. Page and W. K. Wootters, *Evolution without evolution: Dynamics described by stationary observables*, Phys. Rev. D **27**, 2885 (1983).
4. V. Giovannetti, S. Lloyd, and L. Maccone, *Quantum time*, Phys. Rev. D **92**, 045033 (2015).
5. A. Connes and C. Rovelli, *Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories*, Class. Quantum Grav. **11**, 2899 (1994).
6. E. P. Wigner, *On the quantum correction for thermodynamic equilibrium*, Phys. Rev. **40**, 749 (1932).
7. A. S. Holevo, *Probabilistic and Statistical Aspects of Quantum Theory* (North-Holland, Amsterdam, 1982).
8. R. Gambini, R. A. Porto, and J. Pullin, *Realistic clocks, universal decoherence, and the black hole information paradox*, Phys. Rev. Lett. **93**, 240401 (2004); G. J. Milburn, *Intrinsic decoherence in quantum mechanics*, Phys. Rev. A **44**, 5401 (1991).
9. E. Moreva, G. Brida, M. Gramegna, V. Giovannetti, L. Maccone, and M. Genovese, *Time from quantum entanglement: An experimental illustration*, Phys. Rev. A **89**, 052122

(2014).